

LLM SFT WITH QLoRA ON FINEVAL

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1 INTRODUCTION

With the development of the financial industry and the wide application of financial knowledge, an intelligent model that can accurately answer financial exam multiple-choice questions is of great significance. This experiment aims to fine-tune specific models to achieve excellent performance in answering financial exam multiple-choice questions, and at the same time explore the impacts of different models, fine-tuning parameter settings, and data collection on the model performance.

2 EXPERIMENT DESIGN

2.1 DATA SOURCE AND DATA PROCESSING

The data is FinEval jointly produced by Ant and SUFE. Through long-term objective research and summary and strict manual screening, 8351 questions of various types, such as multiple-choice questions, subjective and objective short-answer questions, reasoning and planning questions, and retrieval and answer questions, are used. These questions are highly consistent with actual application scenarios and cover financial academic knowledge, financial industry knowledge, financial security knowledge, and financial agents. Currently, the open-source dataset is financial academic knowledge. FinEval financial academic knowledge is a collection of high-quality multiple-choice questions, covering fields such as finance, economics, accounting, and certificates. It includes 1321 training sets with answers, covering 34 different academic subjects. Compared with the link provided in the assignment, I found the original repository and added Chinese questions made by Ant, increasing the number of questions to 1781. Also, the dev folder of SUFE contains a small amount of dev data with COT, and each subject has 5 answers with COT. At the same time, for general knowledge questions, I selected CHARM, covering Chinese common knowledges.

Format Conversion and Dataset Partitioning The original data is converted into a specific shareppt-formating JSON file for model processing. The dataset is divided into a training set and a test set to ensure the independence and effectiveness of the training and evaluation processes.

COT Data Integration In the dev training set, COT data with explanations (explanation) is added. It is expected that these explanatory information can help the model better understand the problem-solving logic and improve its answering ability.

CHARM Data Introduction Multiple-choice questions of general-domain knowledge CHARM are incorporated to maintain the balance between domain-specific data and general data, ensuring that the model has domain-specific capabilities while maintaining generality.

Data Augmentation Data augmentation is carried out on the prompt instructions. Through diversified instruction expression methods, the model is made more robust to various question-asking forms, improving the model's adaptability in practical applications.

2.2 MODEL SELECTION

Untuned Models First, test the performance of untuned models on the test set as a benchmark for comparing the performance of fine-tuned models later. This helps to clarify the actual contribution of the fine-tuning process to the improvement of model performance.

Fine-Tuned Models Two models, Qwen2-7b-instruct and DeepSeek-R1-Distill-Qwen-7B, are selected for fine-tuning research. Qwen2-7b-instruct has been fine-tuned with a general-purpose dataset and can adapt to the instructions of question-answering questions. DeepSeek-R1-Distill-Qwen-7B uses the Chinese reasoning dataset generated from DeepSeek to fine-tune Qwen-7B, has reasoning ability, and has excellent performance in Chinese context and domain-knowledge understanding. Such a comparative experiment can study the performance of reasoning large models and instruction large models in dealing with Chinese-text-based financial exam questions.

2.3 PARAMETER TUNING

Learning Rate Adjustment Two learning rates, $1e-5$ and $2e-7$, are set for the experiment. The learning rate affects the step size of parameter updates during the model training process. An appropriate learning rate can enable the model to converge to a better performance state faster.

Exploration of Training Epochs Training for 1 epoch and 5 epochs is attempted respectively. The number of training epochs determines the number of times the model learns from the training data. Theoretically, a larger number of training epochs can allow the model to better fit the training data, but it may also lead to overfitting. Based on experience, it is speculated that 5-epoch training may bring better results, but it needs to be verified through experiments.

Other important parameters Due to time constraints, further experiments were not carried out. For example, the lora rank was fixed at 8, and larger values can be tried to increase the parameter changes of the SFT dataset on the model.

2.4 FINE-TUNING METHOD

We adopt QLoRA fine-tuning. Based on the quantized 8-bit model, a low-rank matrix is used to adjust some parameters of the model. While reducing the computational cost and memory requirements, the model can be effectively fine-tuned. Compared with full-parameter fine-tuning, LoRA fine-tuning is more feasible in resource-limited situations, has a shorter training time, and requires less GPU memory, and has been widely used in multiple experiments. Other fine-tuning methods, such as the AdamW optimizer and the P-tuning method, were not further explored due to time constraints.

3 CODE IMPLEMENTATION

3.1 SINGLE-CARD TRAINING AND INFERENCE CODE BASED ON TRANSFORMERS

see NLP-hw3.ipynb

3.2 MULTI-CARD TRAINING CODE BASED ON LLAMAFACTORY

see NLP-hw3-llamafactory.ipynb

4 RESULT ANALYSIS

model name	dataset	epoch	lr	time	last loss	fin acc w/o SFT	fin acc w/ SFT	CHARM acc w/ SFT
Qwen2	fineval	1	$2e-7$	1h	5.805	70.03%	81.23%	67.78%
Qwen2	+cot+charm	1	$2e-7$	1h12m	14.141	70.03%	80.39%	65.00%
Qwen	+cot+charm	5	$2e-7$	6h12m	13.457	70.03%	81.23%	65.00%
Qwen2	+cot+charm	1	$1e-5$	1h26m	8.408	70.03%	85.46%	68.33%
DeepSeek	fineval	1	$2e-7$	1h21m	17.997	48.18%	48.79%	32.78%

Table 1: Experimental Results

An example response for qwen2-7b-instruct

"query": "请回答下面的单选题，请直接正确答案选项，不要输出其他内容。以下是中国常识的单项选择题，请选出其中的正确答案。和这些电影《春光乍泄》、《旺角卡门》、《一代宗师》、《堕落天使》有共同点的电影是：选项：(A)《枪火》(B)《任逍遥》(C)《山河故人》(D)《花样年华》”，
 "ground truth": "D",
 "model answer": "(D)《花样年华》”

Figure 1: An example response for qwen2-7b-instruct

An example response for Deepseek

"query": "请回答下面的单选题，请直接正确答案选项，不要输出其他内容。以下是中国常识的单项选择题，请选出其中的正确答案。和这些电影《春光乍泄》、《旺角卡门》、《一代宗师》、《堕落天使》有共同点的电影是：选项：(A)《枪火》(B)《任逍遥》(C)《山河故人》(D)《花样年华》”，
 "ground truth": "D",
 "model answer": "嗯，我现在得回答这个问题。题目是关于中国电影的，四个选项分别是(A)《枪火》、(B)《任逍遥》、(C)《山河故人》、(D)《花样年华》。问题是要找出与《春光乍泄》、《旺角卡门》、《一代宗师》、《堕落天使》这四部电影有共同点的电影。首先，我需要分析这四部电影的共同点是什么。《春光乍泄》是由冯小刚执导的，讲述了一个女性追求自由的故事，可能涉及一些社会议题。《旺角卡门》也是冯小刚的电影，通常这类电影可能探讨爱情、生活和 maybe 一些社会现象。《一代宗师》则是贾樟柯导演的，可能更偏向于文艺、maybe 有深刻的叙事或主题。《堕落天使》是冯德伦导演的，可能涉及更个人化的故事，比如爱情、maybe 人生选择。接下来，我得考虑这些电影的共同点。可能它们都涉及到爱情、个人成长、maybe 一些社会议题，或者导演的不同风格。比如，冯小刚和贾樟柯的电影可能在叙事”，
 "model choice": "D”

Figure 2: An example response for Deepseek

4.1 RESPONSE CASES

The qwen2-7b-instruct model directly outputs the results. On the CHARM dataset, there may be cases of extra parentheses and option content, while on the financial dataset, it directly outputs the option letters. For example:

Deepseek’s reasoning is too long to be completed within max new token. Even if it is written in the query to directly output the option result, it cannot follow the instruction. This causes the evaluator to determine the last option mentioned in its thinking as its answer. For example:

However, when the answer is successfully selected within the maximum limit of 512 tokens, deepseek will write the answer after , for example:

Due to the overly long reasoning, the overall accuracy of Deepseek is very low, which implies that reasoning ability and instruction-following ability may be mutually exclusive.

4.2 IMPACT ANALYSIS OF EXPERIMENTAL COMPONENTS ON MODEL PERFORMANCE

Impact of Model Selection From the preliminary results, different models show significant performance differences under the same fine-tuning configuration. The Qwen2-7b-instruct series of models show relatively high accuracy under some configurations, while the DeepSeek-R1-Distill-Qwen-7B model has relatively low accuracy before and after SFT under the same QLoRA epoch = 1, lr = 2e-7 configuration. This implies that reasoning ability and instruction-following ability may be mutually exclusive.

Impact of Epoch In the Qwen2-7b-instruct model, when the number of training epochs increases from epoch = 1 to epoch = 5, the loss decreases slightly, but the curve shows a significant downward trend, and the accuracy on the fineval dataset increases by 1%.

Another example response for Deepseek

"query": "请回答下面的单选题，请直接正确答案选项，不要输出其他内容。以下是中国金融考试的单项选择题，请选出其中的正确答案。根据《期货公司监督管理办法》的规定，期货公司分支机构终止的，应当先行妥善处理该分支机构的客户资产，结清并终止经营活动。A:工作人员工资 B:分支机构业务 C:交易账户和客户编码 D:营业场所和设施的相关费用”，

"ground truth": "B”，

"model answer": "嗯，我现在要解决这个关于中国金融考试的单选题。题目是关于期货公司分支机构终止的相关规定，选项有四个：A是工作人员工资，B是分支机构业务，C是交易账户和客户编码，D是营业场所和设施的相关费用。首先，我需要回忆一下《期货公司监督管理办法》的相关内容。我记得，关于分支机构终止的规定，主要是要妥善处理客户资产，结清业务，然后终止经营活动。其中，结清业务可能包括一些具体的东西，比如交易账户和客户编码，因为这些是直接关系到客户的交易记录和信息。让我再仔细想想，分支机构终止的时候，通常需要先处理客户的资产，比如资金、证券等。然后，结清业务，可能包括交易账户和客户编码，这样确保客户不会因为业务终止而受到影响。同时，可能还要支付工作人员的工资，以及营业场所和设施的相关费用，这些都是终止的一部分，但题目问的是“先行妥善处理客户资产，结清并终止经营活动”。所以，正确的选项应该是结清分支机构业务，也就是选项B。因为结清业务是终止的重要步骤，确保业务正常收尾，避免遗留问题。再确认一下，A是工作人员工资，这可能是在终止时需要支付的费用，但题目问的是“结清”，所以可能不是。C是交易账户和客户编码，这可能也是结清的一部分，但通常结清业务可能更广泛，包括各种业务的结算。而D是营业场所和设施的费用，这可能是在终止时需要支付的费用，但同样，题目问的是“结清”。所以，综合来看，选项B是正确的，因为结清分支机构业务是终止过程中的关键步骤，确保业务顺利收尾。;/think_答案: B”，

"modelchoice": "B”

Figure 3: Another example response for Deepseek

Impact of Learning Rate In the Qwen2-7b-instruct model, when the learning rate increases from $2e-7$ to $1e-5$, the loss decreases significantly, and the accuracy is up to 85.46%

Impact of Data Augmentation Adding COT data with explanations and multiple-choice questions of general-domain knowledge CHARM did not show obvious advantages in this experiment.

5 CONCLUSION

This experiment focuses on the task of answering financial exam multiple-choice questions. Through careful fine-tuning of specific models, it deeply explores the impacts of different models, parameter settings, and dataset construction on the model performance.

Among the two models, Qwen2-7b-instruct and DeepSeek-R1-Distill-Qwen-7B, Qwen2-7b-instruct shows relatively high accuracy under some configurations. On the fineval dataset, under the QLoRA epoch = 1, lr = $2e-7$ configuration, the eval accuracy reaches 70.03%, and after subsequent configuration adjustments, it can reach up to 85.46%. While DeepSeek-R1-Distill-Qwen-7B, under the same configuration, has an accuracy of only 48.18% before SFT and 48.79% on the financial dataset after fine-tuning, with significantly inferior performance. In terms of instruction-following, the Qwen2-7b-instruct model can directly output option letters on the financial dataset, but there are cases of incorrect output formats on the CHARM dataset. The DeepSeek-R1-Distill-Qwen-7B model has difficulty following instructions due to its overly long reasoning process, suggesting that there may be a mutual-exclusive relationship between reasoning ability and instruction-following ability.

In the parameter-tuning experiments, it is found that a larger epoch can improve the model performance, and an increase in the learning rate can significantly reduce the loss and improve the model performance. In terms of dataset construction, adding COT data and general-domain knowledge did not show obvious advantages in improving the model performance in this experiment.

This experiment has successfully constructed a research system for fine-tuning models for financial exam multiple-choice questions, clearly presenting the performance differences of different models in this task, as well as the impacts of parameter tuning and data processing methods on the model performance. Although there is still room for improvement in model performance, instruction-following, and data utilization, it provides valuable experience and clear directions for follow-up research. Future research can focus on optimizing model selection and fine-tuning strategies, deeply exploring the balance between reasoning ability and instruction-following ability, and further tapping the potential of data augmentation and new data introduction to achieve a breakthrough in the performance of models in answering financial multi-choice questions.

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REFERENCES